

TLabel: A Python library for standardized tactile data annotation and quality assessment

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Abstract

Tactile sensors from different manufacturers produce incompatible data formats, and current annotation tooling provides no standardized vocabulary for cross-sensor comparison. Unavailable measurements are frequently conflated with zero or estimated values, introducing silent semantic errors into downstream learning pipelines. We describe **TLabel**, an open-source Python library that addresses this through a capability-aware 14-dimensional semantic schema built on a non-fabrication principle: a field is populated only when supported by sensor evidence, otherwise it remains explicitly **None**. The software provides a plug-in adapter system supporting 10 sensor formats, a four-level compliance hierarchy (L1–L4) for automatic quality assessment, and a semantic maturity framework for field-level provenance tracking. Validation on 15,677 real frames from three public datasets confirms 100% boundary compliance. TLabel is available via PyPI (`pip install tlabel`), with source on GitHub and archival on Zenodo.

Keywords: tactile sensing, data annotation, semantic schema, quality assessment, Python, robot learning

Metadata

1. Motivation and significance

Robotic manipulation relies on tactile feedback for contact detection, force estimation, and slip identification [1, 2]. As sensor variety grows—impedance-based electrodes [3], vision-based elastomers [2], piezoresistive taxel grids [4]—so does data heterogeneity. While large-scale robot learning initiatives have standardized vision and proprioception (Open X-Embodiment [5])

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Nr.	Code metadata description	Metadata
C1	Current code version	v0.21.1
C2	Permanent link to code/repository used for this code version	https://github.com/liesliy/tlabel/releases/tag/v0.21.1
C4	Legal Code License	Apache-2.0
C5	Code versioning system used	Git
C6	Software code languages, tools, and services used	Python 3, numpy, pandas; optional: opencv-python, h5py, pyarrow
C7	Compilation requirements, operating environments	Python ≥ 3.8 ; Linux, macOS, Windows
C8	If available Link to developer documentation/manual	https://github.com/liesliy/tlabel
C9	Support email for questions	luoxi@touchlabelai.cn

Table 1: Code metadata (mandatory).

in RLDS [6], LeRobot [7, 8]), neither prescribes tactile semantic fields nor how missing measurements are represented.

A fundamental failure mode arises when *unavailable* measurements are conflated with *zero* or *estimated*. When a schema fills unsupported fields with defaults, downstream models silently convert format incompatibility into false observations. TLabel addresses this as a *semantic contract*: a field is populated only when supported by sensor evidence; otherwise it remains explicitly **None**.

2. Software description

TLabel organizes tactile annotation around three layers: (i) raw data input from heterogeneous sensors (CSV, PKL, H5, NPY, Parquet, SDK streams); (ii) an adapter layer implementing capability-aware parsing and semantic routing; and (iii) a unified 14-dimensional schema with multi-format export (JSON, CSV, Zarr, LeRobot).

2.1. Core data model

The central data structure is a 14-dimensional typed **dataclass** spanning five groups: **Contact** (binary indicator, optional 2D centroid), **Force** (scalar magnitude, optional 3D vector), **Slip** (binary event, optional manipulation phase), **Surface** (deformation, temperature), and **Meta** (confidence, compliance level). Each dimension has a provenance classification (*Direct/Derived/Reserved*) and maturity label (*Stable/Tentative/Experimental/Reserved*).

Four dimensions are *Reserved*, providing forward-compatible slots for future modalities.

Each adapter declares a static capability set $C_A \subseteq \mathcal{S}$. For any frame f : $\forall d \in \mathcal{S} \setminus C_A : f.d = \text{None}$, enabling consumers to filter sensors by required capabilities without inspecting individual frames.

2.2. Compliance and maturity

Four compliance levels are assigned automatically: **L1** (binary contact only), **L2** (contact plus quantitative measurement), **L3** (L2 plus spatial information), and **L4** (L3 plus multi-modal coverage). The maturity framework classifies each field by *mapping type* and *validation evidence*: **temperature** from a thermistor is *Stable/Direct*; **slip_event** from a heuristic is *Tentative/Derived*; **texture_class** with no adapter is *Reserved*.

2.3. Plug-in adapter system

TLabel employs a dual-base-class architecture: **DataAdapterBase** for recorded files, **SensorAdapterBase** for live hardware streams. Both follow an inheritance-based pattern—(i) inherit, (ii) implement `extract()`, (iii) register for automatic discovery—with the base class enforcing `compliance_level` assignment.

TLabel ships with 10 built-in adapter implementations covering sensor formats from GelSight Mini, DIGIT, PaXini (PXCcap and GEN3), Daimon (DM-Tac and DM-TacClaw), UniVTAC, TacQuad, VTouch, YCB-Slide, and TouchD. Third-party adapters use the `tlabel` entry point for automatic discovery.

2.4. Export and integration

Annotated frames carry full provenance metadata. Export targets include JSON, CSV, FTP-1 Zarr (for foundation model training), and LeRobot dataset format.

3. Illustrative examples

We demonstrate TLabel using three real-world public datasets (Table 2).

Example 1: Boundary compliance. For each dataset, we verify that fields outside the adapter’s capability set C_A are **None** for all frames. We define the boundary compliance rate B as the fraction of unsupported field assignments that are correctly **None**. Across all 15,677 frames, $B = 100\%$: BioTac sets 8 fields to **None** per frame (no spatial resolution); YCB-Slide degrades to L1 with 8/14 fields **None**; Sparsh populates 8/14 fields including full 3D force vectors matching ATI ground truth at 100%.

Table 2: Dataset characteristics and boundary verification results.

Dataset	Level	Frames	Supported	Unsupported	Violations
BioTac BTS3 [9]	L2	1,276	6/14	8/14	0
YCB-Slide [10]	L1	3,661	6/14	8/14	0
Sparsh [11]	L3	10,740	8/14	6/14	0
Total		15,677			0

```
import tlabel

# Load any sensor (format auto-detected)
data = tlabel.load("sparsh_trajectory/")
print(data.compliance_level) # "L3"
print(data.schema.supported) # 8/14 fields

# Export to downstream formats
data.export("output.json") # full provenance
data.export_ftpl("out.zarr") # foundation model ready
```

Example 2: Semantic corruption detection. Using 10 features from the Sparsh dataset with field-missing rates of 0%, 42%, 59%, and 78% (c0–c3, respectively), we compare: **A** (naive zero-fill), **B** (zero-fill + observation masks), and **C** (TLabel-equivalent: B + maturity one-hot). For logistic regression, zero-fill degrades from area under the precision–recall curve (AUPRC) 1.000 to 0.531, while observation masks preserve 0.607—a gain of +0.05 to +0.11. Tree-based models remain robust (AUPRC ≥ 0.967), showing that explicit missingness benefits are model-dependent.

4. Impact

TLabel fills a gap in embodied AI data infrastructure as the first open-source, sensor-agnostic semantic standard for tactile data.

Reproducibility. Explicit missingness and full provenance tracking ensure datasets can be audited. The 100% boundary compliance across 15,677 frames confirms the non-fabrication principle as a structural invariant.

Interoperability. The plug-in adapter system and multi-format export (JSON, CSV, Zarr, LeRobot) enable integration with existing pipelines, with community contributions supported through the entry-point architecture.

Downstream reliability. Naive zero-filling degrades linear models by up to 0.11 AUPRC; TLabel’s explicit missingness provides a principled alternative.

TLabel (248 commits, 42 stars) is available via PyPI [12] (`pip install tlabel`), GitHub (<https://github.com/liesliy/tlabel>), and Zenodo [13].

5. Conclusions

We have described TLabel, an open-source Python library implementing a capability-aware 14-dimensional semantic schema with explicit missingness, a four-level compliance hierarchy, and a plug-in adapter system supporting 10 sensor formats. Validation on three public datasets confirms 100% boundary compliance. Future work includes additional adapters, deeper LeRobot integration, and alignment with ISO 26264-1.

Data Availability Statement

All datasets used in this study are publicly available. BioTac BTS3 from [9]; YCB-Slide from [10]; Sparsh from [11]. No new data were generated.

Declaration of Competing Interest

The author declares no competing interests.

CRedit Author Statement

Xi Luo: Conceptualization, Software, Methodology, Validation, Writing – original draft, Writing – review & editing.

Acknowledgments

All code and data processing were conducted at the Tactile Intelligence Lab, Niuxiu Technology. The author thanks the open-source community for contributions to the tlabel package, including the external pull request from gaolebaigao.

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